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## Digital Passion

Fuelling the pursuit of technology knowledge

Lead Feature

# The Final Frontier

Billions of dollars spent on space research might not have taken us any further than our own Moon. But all that's about to change... Let's take a look at how research in this field not only helps us travel to distant planets, but also improves things right here on Earth

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**H**umans are, by nature, inquisitive. We want to learn, we want to invent, and most of all, we want to explore all possibilities. It's this very quality of homo sapiens that has made us evolve from little tribes living on the banks of rivers—wielding sticks, worshipping fire and battling the elements—to a six billion-strong society with the monstrous concrete jungles we call home, and the ability to fly off this planet! But why should we ever leave Earth?

### Problems

We have explored the majority of our planet, and exploited it in every way. What happens when we run out? Of fuel to burn, or real estate to live in? We can't manufacture oil; we haven't found a replacement for it; we can't, or won't, even build cities underground, or underwater. We can't halt the population explosion either. If only there were another Earth!

### Solutions

Yes, there are hundreds of thousands of suns, billions of planets, and a myriad of moons out there in space. There is life out there, and perhaps even intelligent life. We're fascinated by the idea of visiting other planets, and would love to meet an alien

being—of the *E.T.* type, not the *Independence Day* or *War Of The Worlds* type!

One popular science-fiction (sci-fi) solution for the fuel problem is to set up mining colonies on other planets, even on the uninhabitable ones, to bring back minerals and fuels to Earth. And we don't mean from galaxies far, far away. We are looking at our neighbour Mars. But we actually know nothing about it. Sure, NASA has sent probes and Mars Rovers over there, but even then, we're just guessing.

The only proper way to understand Mars is to go there ourselves, live there for a while, drill for minerals, explore... However, the first hurdle we hit when we think of space travel, or more appropriately space exploration, is, how do we get there? As with many things in life, we need technology to help us do this.

The human body is fragile, and needs some very basic conditions to be met in order to survive—whether in space or on Earth. It's as simple as providing air, food and water. But when it comes to space even this is difficult.

In order to be able to travel in space, we need a lot more than engines and huge space cruisers. We need to have a plentiful supply of water, food, and oxygen, and we need to find ways to have enough of it on board our space vessels, with the ability to replenish our supplies. But then again, sending huge loads of supplies into space require better and more efficient rocket propulsion systems... the problems are endless!

When it comes to the computing technology required to explore space, it's obvious that we need really good computing machines, capable of making gargantuan calculations and accounting for millions of variables with incredible precision. When we talk in terms of